



GE VERNOVA

ADMS

16.25.1

DNOM2HDB User Guide

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Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
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1. Introduction

The DNOM2HDB application copies data from DNOM to a Habitat database and to Historian. The following image illustrates the DNOM2HDB application role in the ADMS archiving process.

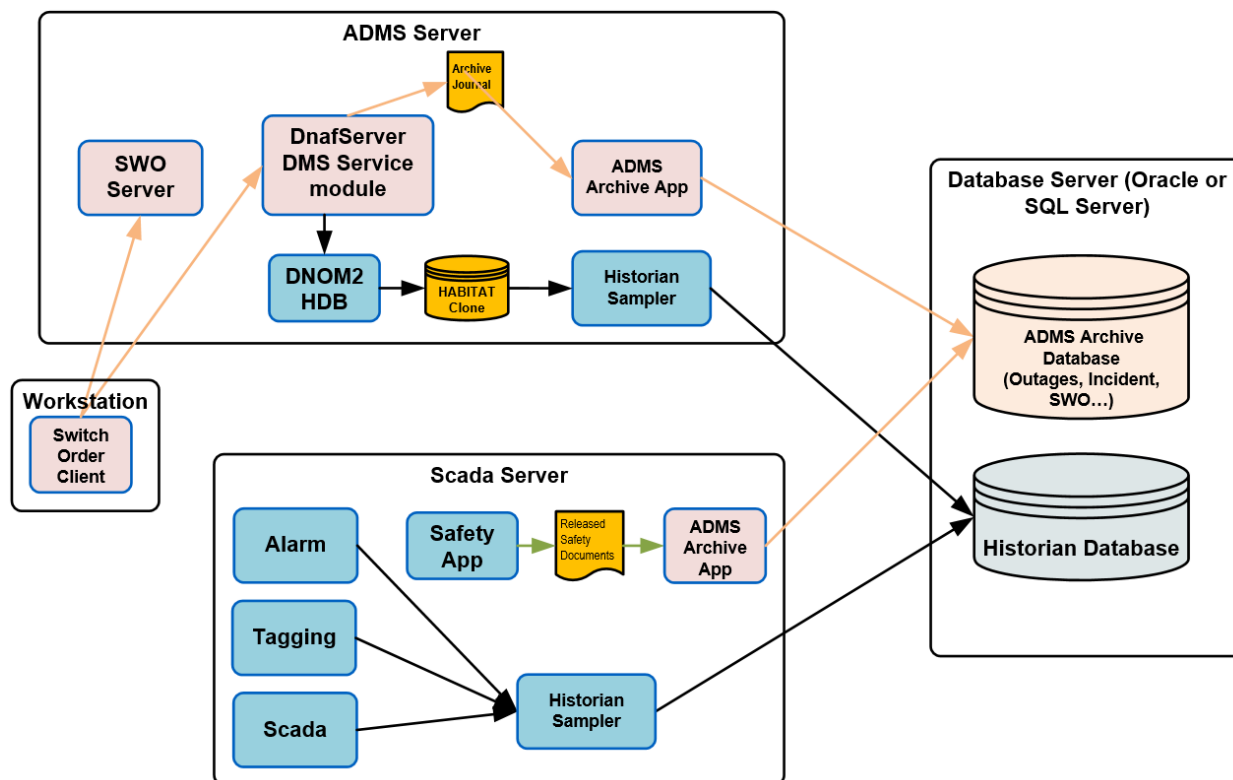


Figure 1. ADMS Archiving Capabilities

However, the main purpose of DNOM2HDB is to copy data from DNOM fields associated with a Query Helper Function or Field Map into any Habitat database (HDB) at any time.

All fields defined in the station model files have their attributes defined in the field map. Most other attributes, including application results, are declared with a query helper function. These query helper functions commonly translate raw data to a more user-friendly format, including translating Boolean values to True/False or Yes/No.

For guidance on how to create a query helper function, refer to the *ADMS Modeling Reference Guide*.

Virtually, DNOM2HDB can copy DNOM data to any HDB clone, as shown in the following image.

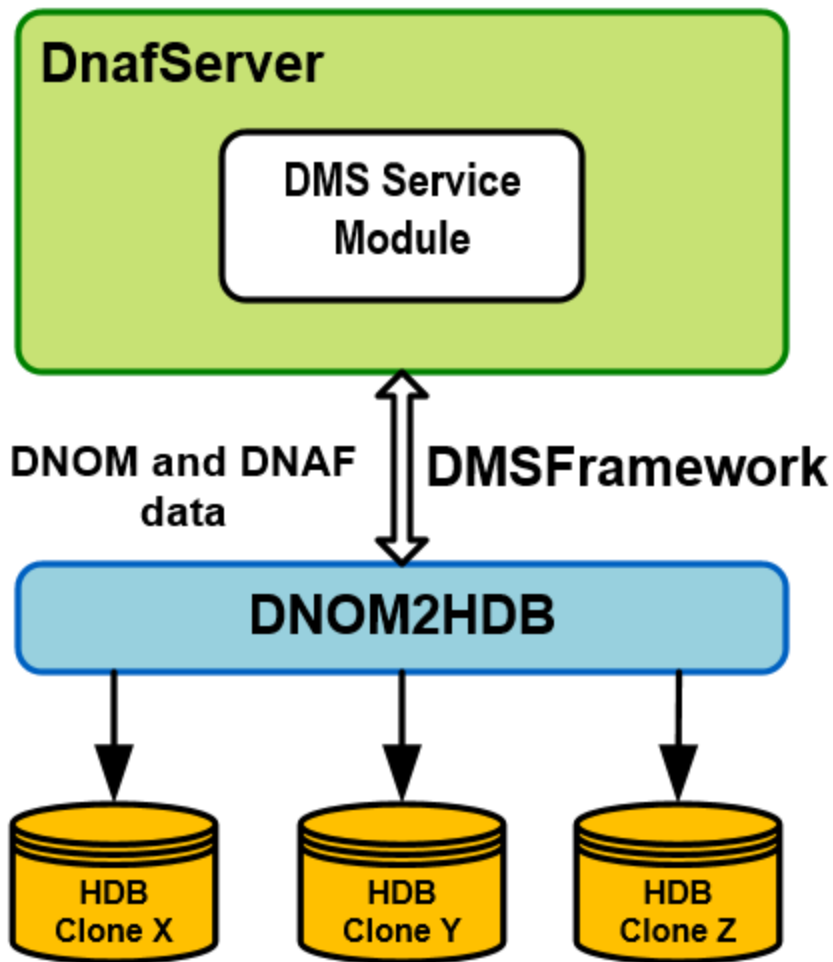


Figure 2. DNOM2HDB Data Flow Overview

1.1. Functional Overview

This section provides a high-level overview of the DNOM2HDB query syntax and execution triggers.

1.1.1. Query Syntax

The DNOM2HDB query syntax is based on the DNOM query syntax defined in the *ADMS Modeling Reference Guide*.

To query for the names and IDs of all switching devices in a "station" that are both a closed breaker and are currently closed, the entire switch query is written as follows:

```
SELECT NAME,ID FROM RT.SUB(stationX).SWT WHERE CLOSEDBREAKER TRUE, CURROPEN 0
```

It is possible to use a virtual object for querying from multiple objects that belong to the same category (for example, to query some fields from all branch devices). Branch devices in one

substation may include lines, switching devices, transformers, reactors, and cuts. In reality, there is no device called "BRANCH" in one substation. To satisfy the query requirement in the above case, the concept of "virtual objects" is supported in the DMS query helper functions. Virtual objects map to multiple real DNOM objects that share some common attributes. In the above case, the virtual object is called "BRANCH", which maps to three real DNOM objects: SWT, CUT, and LN. You can write the query statement using the following format:

```
SELECT NAME, ID FROM RT.SUB(stationX).BRANCH
```

When the query statement is executed against one virtual object, it is executed against all mapped real DNOM objects.

If the virtual object "BRANCH" has one field called "X", you can also execute the following statement to query the field "X" from the actual "LN" object:

```
SELECT X FROM RT.SUB(stationX).LN
```

For a list of all available virtual object names and their mapped real DNOM object names, refer to the "Virtual Objects" worksheet in the *ADMS Modeling Reference Tables*.

For more information about how to edit query and record settings, see [Configuring DNOM2HDB](#).

1.1.2. Query Trigger

DNOM2HDB executes on every "heartbeat" - a periodic treatment time. At each heartbeat, the queries are evaluated to determine if they can be executed. The periodicity of the heartbeat is configurable on the Parameters display.

A query can be triggered on user demand or from a timer:

- On user demand, the query is immediately executed.
- On timer, the query may be re-executed on a station if at least some specific amount of time has elapsed since the last execution on this station.

It is possible to add extra constraints on the timer execution. The constraint can be on topology change or on the execution state of a DNAF application. In this case, DNOM2HDB delays the query execution until the topology has changed in the station, or until the DNAF application has reached a specific execution status for the station.

DNOM2HDB allows execution status constraints for these DNAF applications:

- **AFR:** Automatic Feeder Reconfiguration.
- **DPF:** Distribution Power Flow.
- **PRV:** Protection Validation.
- **FISR:** Fault Isolation and Service Restoration.
- **LVM:** Load and Voltage/VAR Management.
- **FL:** Fault Location.

For details about how to set the execution constraints, see [Configuring DNOM2HDB](#).

For example, if a user sets a query based on a timer and a constraint on DPF requiring the Converged status, the query is executed on two DMS stations: A and B. The timer limit is set to four minutes. The heartbeat periodicity is set to 1 minute.

- On $t=0$ the query is executed on both station A and B.
- On $t=4$ the timer limit has expired, the query is executed on station B only. The query is not executed on station A because the DPF has not been recalculated since $t=0$.
- On $t=8$ the query is not executed on station B because the DPF status is not "converged".
- On $t=9$ the query is executed on station B because the DPF is "converged".
- On $t=10$ the query is executed on station A because the DPF has "converged".

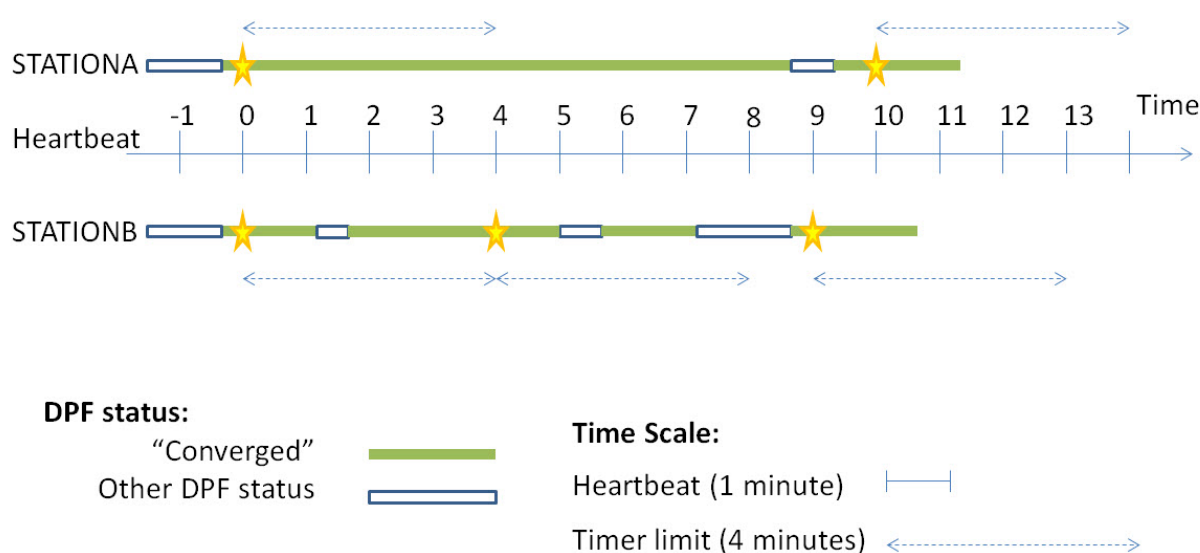


Figure 3. Example Query Timer

1.1.3. Preselected Station List

You can enter the list of stations related to a query, or you can associate a preselected station list.

This preselected station list is unique in the DNOM2HDB application. For more information about how to configure the list of stations, see [Configuring DNOM2HDB](#).

2. Configuring DNOM2HDB

The following sections describe the two steps for configuring a DNOM-HDB mapping:

1. Enter the HABITAT record reference to receive the data.
2. Define a DNOM Query request which will identify the DNOM data to read and the mapping to the HDB fields where the data will be written.

2.1. Configuring the HABITAT Record

The HDB Tables display lists the user-entered HDB records.

To access the HDB Tables display, from the Browser menu, navigate to HABITAT Applications > DMS2HDB > HDB Tables.

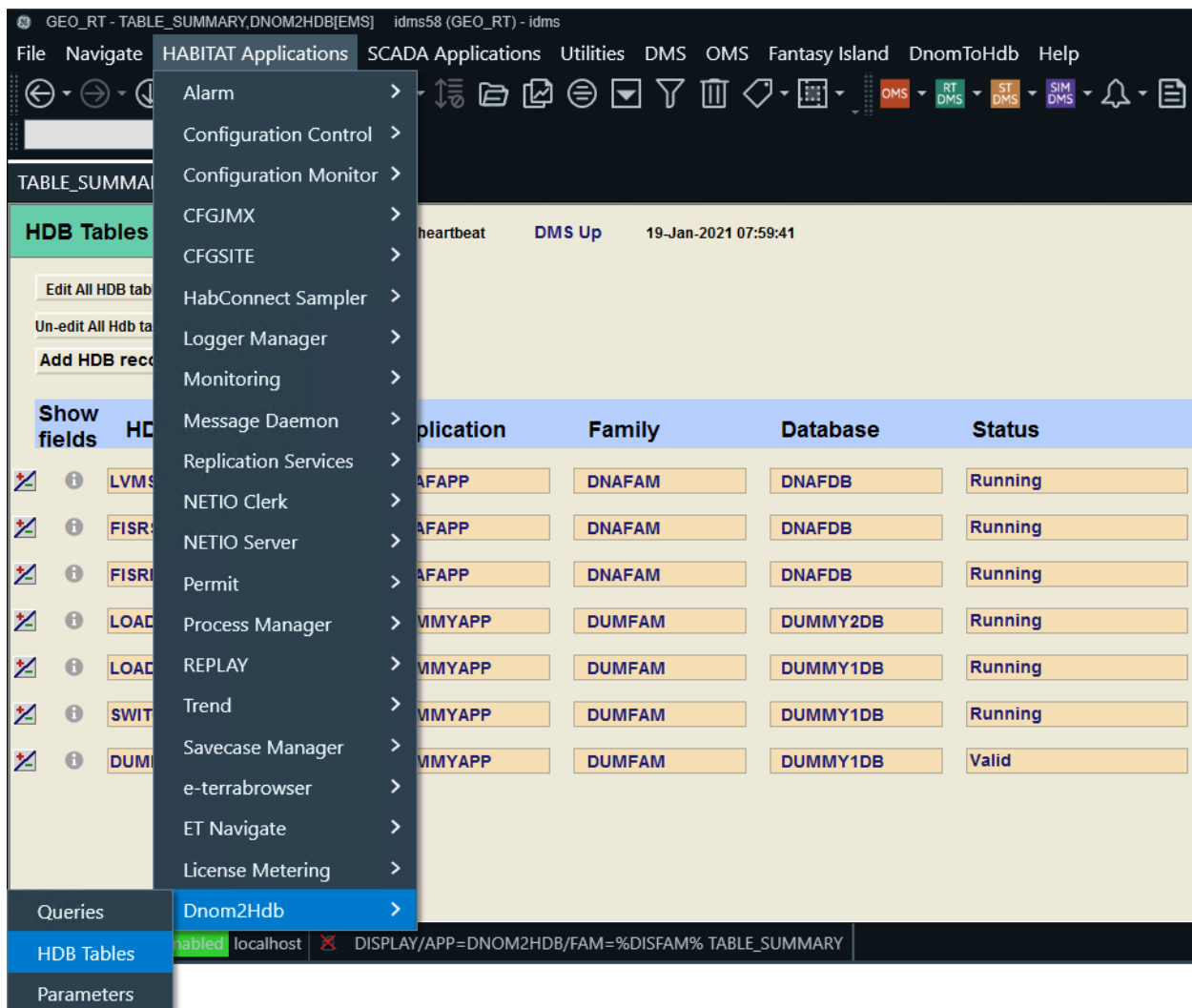


Figure 4. HDB Tables Display

2.1.1. Editing HDB Tables

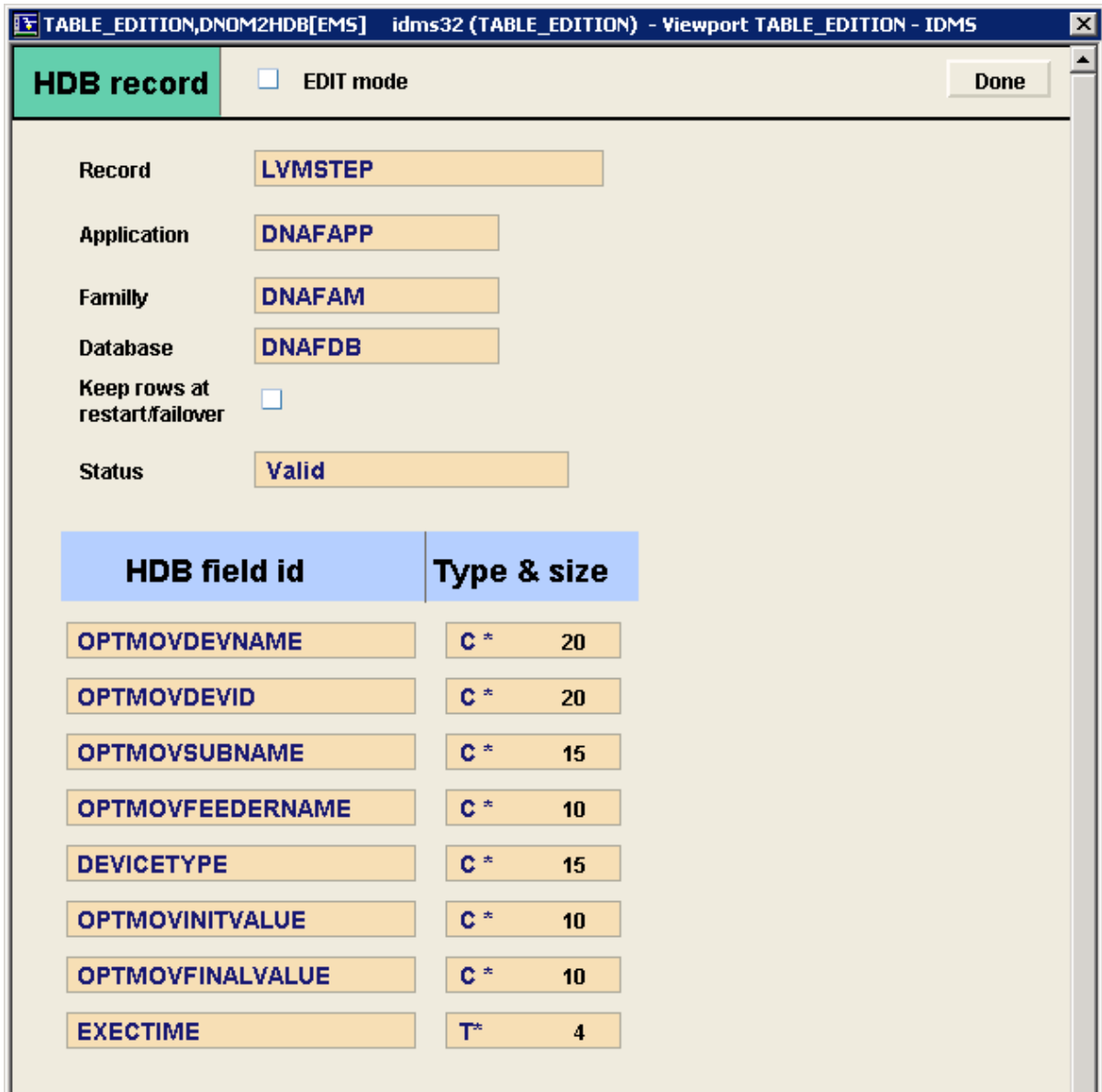
To add or remove a HABITAT record configuration:

- Right-click the "+/-" button and choose an option.
- Click Add HDB Record.

To edit the existing HDB record configuration:

1. Click on the information icon  for the HDB record to be edited.

The HDB Record pop-up display appears.



HDB field id	Type & size
OPTMOVDEVNAME	C * 20
OPTMOVDEVID	C * 20
OPTMOVSUBNAME	C * 15
OPTMOVFEEDERNAME	C * 10
DEVICETYPE	C * 15
OPTMOVINITVALUE	C * 10
OPTMOVFINALVALUE	C * 10
EXECTIME	T* 4

The HDB Record display contains the read-only information that is automatically defined by DNOM2HDB once the operator exits Edit mode.

2. Select the Edit check box to enter Edit mode, so you can modify the record settings.

When the Edit check box is cleared, DNOM2HDB verifies that the HDB mapping is correct and updates the status field accordingly.

3. Modify the following information as needed: HDB record name, Application name, Family name, Database name.
4. Select the Keep Rows at Restart/Failover option to keep any existing rows of the HABITAT table when DNOM2HDB is starting or during a failover.

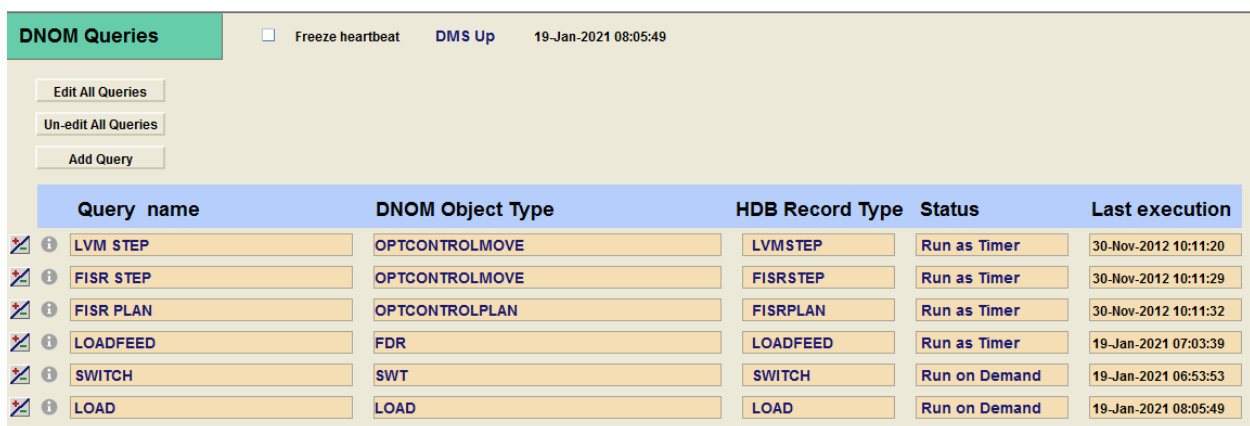
Other fields of the HDB Tables display include:

- Status: Displays current status of the HDB mapping, including:
 - Error: Not initialized
 - Mapping is good
 - Editing mode
- Last update: Time of the last update in the HDB table.
- Information icon: When clicked, opens the HDB Record display.

2.2. Configuring DNOM Query

The DNOM Queries display lists the configured DNOM Query requests and allows the user to insert or delete a DNOM query.

To access the DNOM Queries display, from the Browser menu, navigate to HABITAT Applications > DMS2HDB > Queries.



	Query name	DNOM Object Type	HDB Record Type	Status	Last execution
	LVM STEP	OPTCONTROLMOVE	LVMSTEP	Run as Timer	30-Nov-2012 10:11:20
	FISR STEP	OPTCONTROLMOVE	FISRSTEP	Run as Timer	30-Nov-2012 10:11:29
	FISR PLAN	OPTCONTROLPLAN	FISRPLAN	Run as Timer	30-Nov-2012 10:11:32
	LOADFEED	FDR	LOADFEED	Run as Timer	19-Jan-2021 07:03:39
	SWITCH	SWT	SWITCH	Run on Demand	19-Jan-2021 06:53:53
	LOAD	LOAD	LOAD	Run on Demand	19-Jan-2021 08:05:49

Figure 5. DNOM Queries Display

From the header of the display, you can freeze the heartbeat, see the DMS Service connection status, and see the last time of a successful connection to the DMS Service.

The following information is displayed:

-

The user-defined query name

- The DNOM object type
- The HDB record type
- The current status of the query
- The last execution time of the query

To edit a DNOM query:

1. Click the information icon for the query you need to edit. The Edit Query display appears.

Edit QUERY

☐ EDIT mode

Status
 Run as Timer

Last execution time for this query
 10-Apr-2015 03:00:03
 Done

Query Name

LVM STEP

FROM

☒ Selected station list
 Number max of stations per DNOM query (0 means no limit)
 0

☐ Custom entered stations

WHERE (condition)

ISVVCMOVE=TRUE

DNOM Object Type

OPTCONTROLMOVE

HDB Record Name

LVMSTEP

Trigger

☐ Disable
 ☐ On demand
 ☒ Periodic
 60 seconds
 ☒ Start time

☒ Stop time

☐ No extra constraint
☐ DPF
☒ DNAF applications
☐ FISR
☐ Topology change
☒ LVM

Skip	Destination		Source				Static value
	HDB field id	Type & size	DNOM field	Trigger time	Present time	Constant value	
☐	OPTMOVDEVNAME	C * 20	☒ OPTMOVDEVNAME	☐	☐	☐	☐
☐	OPTMOVDEVID	C * 20	☒ OPTMOVDEVID	☐	☐	☐	☐
☐	OPTMOVSUBNAME	C * 15	☒ OPTMOVSUBNAME	☐	☐	☐	☐
☐	OPTMOVFEEDERNAME	C * 10	☒ OPTMOVFEEDERNAME	☐	☐	☐	☐
☐	DEVICETYPE	C * 15	☒ DEVICETYPE	☐	☐	☐	☐
☐	OPTMOVINITVALUE	C * 10	☒ OPTMOVINITVALUE	☐	☐	☐	☐
☐	OPTMOVFINALVALUE	C * 10	☒ OPTMOVFINALVALUE	☐	☐	☐	☐
☐	EXECTIME	T 4	☐	☒	☐	☐	

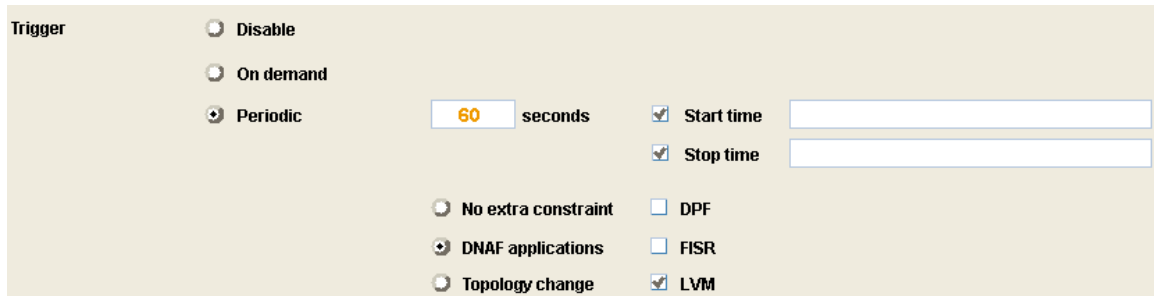
The Edit Query display allows you to view or edit a DNOM query.

The header section of the Edit Query display contains the following information:

- **Status:** Displays the status of the query, including "Editing", "Validated", and "Running".
- **Last Execution Time:** Displays the time of the last successful data copy.

2. To enable editing, select the Edit check box. Clear the Edit check box to validate your updates.

3. Enter any text for identifying the query in the Query Name field.
4. Define the DNOM data according to the DMS Framework Query syntax:
 - For the FROM statement, choose to perform the query on the preselected stations or manually enter the list of stations (see "Custom entered stations" option).
 - Improve performance using the "Number max of station per DNOM query" parameter. For example, enter 5 to split the query into several queries containing a maximum of five stations.
 - Enter constraints in the "WHERE" condition to apply on the query target. For information about the condition syntax, refer to the *ADMS Modeling Reference Guide*.
 - "DNOM Object Type" indicates the DNOM record ID. For example, enter "SWT" for a switch device.
 - "HDB record Name" shows the HDB table to receive the DNOM data. Click the "Select HDB record" button (appears in edit mode) and select the HDB record from the popup. Click the information icon to see the HDB record description in the TABLE_EDITION display.
5. Select one of the following Trigger options:



Trigger

☐ Disable

☐ On demand

☒ Periodic seconds

☒ Start time

☒ Stop time

☐ No extra constraint

☒ DNAF applications

☐ Topology change

☐ DPF

☐ FISR

☒ LVM

- Set the periodic execution time in seconds. This helps to reduce the data volume copied between DNOM to HDB and also allows copying the data only when the DNAF application results are in a correct state.
 - Add some extra constraints, including topology change, or a specific state of some DNAF applications. The expected DNAF status is defined in the Parameters display (see [Configuring General Settings on Parameters Display](#)). Selecting the Topology Change option voids the data copy if the topology has not changed since the previous query execution.
 - Specify a start time or a stop time.
6. Edit any information in the grid at the bottom of the Edit Query display, which contains the list of fields for the selected HDB Record. HDB fields are defined by the user via the HDB Record display.

Skip	Destination		Source				Static value
	HDB field id	Type & size	DNOM field	Trigger time	Present time	Constant value	
☐	OPTMOVDEVNAME	C * 20	☐ OPTMOVDEVNAME	☐	☐		☐
☐	OPTMOVDEVID	C * 20	☐	☐	☐	☐ AAAA	
☐	OPTMOVSUBNAME	C * 15	☐ OPTMOVSUBNAME	☐	☐		☐
☐	OPTMOVFEEDERNAME	C * 10	☐ OPTMOVFEEDERNAME	☐	☐		☐
☒	DEVICETYPE	C * 15	☐ DEVICETYPE	☐	☐		☐
☐	OPTMOVINITVALUE	C * 10	☐ OPTMOVINITVALUE	☐	☐		☐
☐	OPTMOVFINALVALUE	C * 10	☐ OPTMOVFINALVALUE	☐	☐		☐
☐	EXECTIME	T* 4	☐	☐	☐		

- Skip: Select this check box to ignore this HDB field.
- DNOM Field: When selected, enter a DNOM field ID in the text box next to the option button to set the mapping between the DNOM and HDB fields.
- Trigger Time: Maps the trigger time in the HDB field, assuming the HDB field type is fitted to store a date.
- Present Time: Maps the execution time of the query to the HDB field, assuming the HDB field is fitted to store a date.
- Constant Value: Allows entering a constant value in the HDB field. When selected, a text box appears to enter the desired constant.
- Static Value: Used when the DNOM data does not change dynamically and does not require periodical sampling. The data is only re-sent if DNOM2HDB restarts or a failover occurs. The purpose of this option is to improve performance.

7. Clear the Edit check box to validate your updates.

2.3. Editing the Selected Station List

DNOM2HDB automatically refreshes the list of DMS substations. The Station List display allows selecting stations for the queries. To access the Selected Stations List display, from the Browser menu, navigate to HABITAT > DMS2HDB > Stations.

Selected Station list

☐ Freeze heartbeat
 DMS Up
 19-Jan-2021 08:11:13

☐ Edit the station selection

Station				
☒	BACHELOR	DPF 01/19/2021 08:04:21 Solved / BLA Converged	AFR Not Solved Unsolved	FISR Not Solved Unsolved
		LVM 01/19/2021 05:52:36 Solved (No Plan)	PRV 01/19/2021 05:44:33 Solved - No Violations	Topology 30-Nov-2012 10:14:54
☒	BETHPAGE	DPF 01/19/2021 08:04:21 Suspect / BLA 28 of 29 Islands Converged	AFR Not Solved Unsolved	FISR 01/19/2021 06:21:37 Solved
		LVM 01/19/2021 06:25:00 Solved	PRV 01/19/2021 06:22:50 Solved - No Violations	Topology 30-Nov-2012 10:14:54
☒	COHO	DPF 01/19/2021 08:05:21 Solved / BLA Converged	AFR Not Solved Unsolved	FISR Not Solved Unsolved
		LVM Not Solved Unsolved	PRV Not Solved Unsolved	Topology 30-Nov-2012 10:14:54

Figure 6. Selected Station List Display

The Selected Station List display contains the following information:

- The DNOM name of the station.
- The last execution time and the last status of each DNAF application for each station.

To modify the station list:

1. Select "Edit the station selection".
2. Select the check box at the left of the Station header to automatically select all known substations. Alternatively, add stations by selecting the check box for individual station records, as needed.

Selected Station list

☐ Freeze heartbeat
 DMS Up
 19-Jan-2021 08:13:43

☒ Edit the station selection

Station				
☒	BACHELOR	DPF 01/19/2021 08:04:21 Solved / BLA Converged	AFR Not Solved Unsolved	FISR Not Solved Unsolved
		LVM 01/19/2021 05:52:36 Solved (No Plan)	PRV 01/19/2021 05:44:33 Solved - No Violations	Topology 30-Nov-2012 10:14:54
☐	BETHPAGE	DPF 01/19/2021 08:04:21 Suspect / BLA 28 of 29 Islands Converged	AFR Not Solved Unsolved	FISR 01/19/2021 06:21:37 Solved
		LVM 01/19/2021 06:25:00 Solved	PRV 01/19/2021 06:22:50 Solved - No Violations	Topology 30-Nov-2012 10:14:54

2.4. Configuring General Settings on Parameters Display

To access the Parameters display, from the Browser menu, navigate to HABITAT Applications > DMS2HDB > Parameters.

The Parameters display allows you to configure general settings, including:

- Heartbeat: Number of seconds for the periodic heartbeat of DNOM2HDB. At each heartbeat, the queries are evaluated to determine if they can be executed.
- DMS Server: Host name, Port, and Timeout time for the DMS Service module on the DNAF Server.
- Query menu for DNAF applications. Enter the list of DNAF applications with their menu label and status text. This setting is used when a query is set on a timer with the DNAF application constraint (see [Configuring DNOM Query](#)).

The screenshot shows the 'Parameters' configuration page. At the top, there is a header bar with a green 'Parameters' tab, a 'Freeze heartbeat' checkbox (unchecked), a 'DMS Up' status indicator, and a timestamp '19-Jan-2021 05:33:10'. Below the header, the settings are organized into sections:

- Heartbeat:** A text input field containing '30' followed by the label 'Seconds'.
- DMS Server #1:** A group of three inputs: 'Host Name' with 'localhost', 'Socket' with '6033', and 'Timeout' with '120' followed by 'Seconds'.
- DMS Server #2:** A group of three inputs: 'Host Name' with 'host2', 'Socket' with '6033', and 'Timeout' with '120' followed by 'Seconds'.
- Query menu for DNAF applications:** A table with three columns: an icon column, a 'Label Text' column, and a 'DNAF Status' column. It contains three rows for 'DPF', 'FISR', and 'LVM', each with a corresponding status value.

	Label Text	DNAF Status
	DPF	Solved / BLA Converged
	FISR	Solved
	LVM	Solved

Figure 7. Parameters Display

3. Running DNOM2HDB

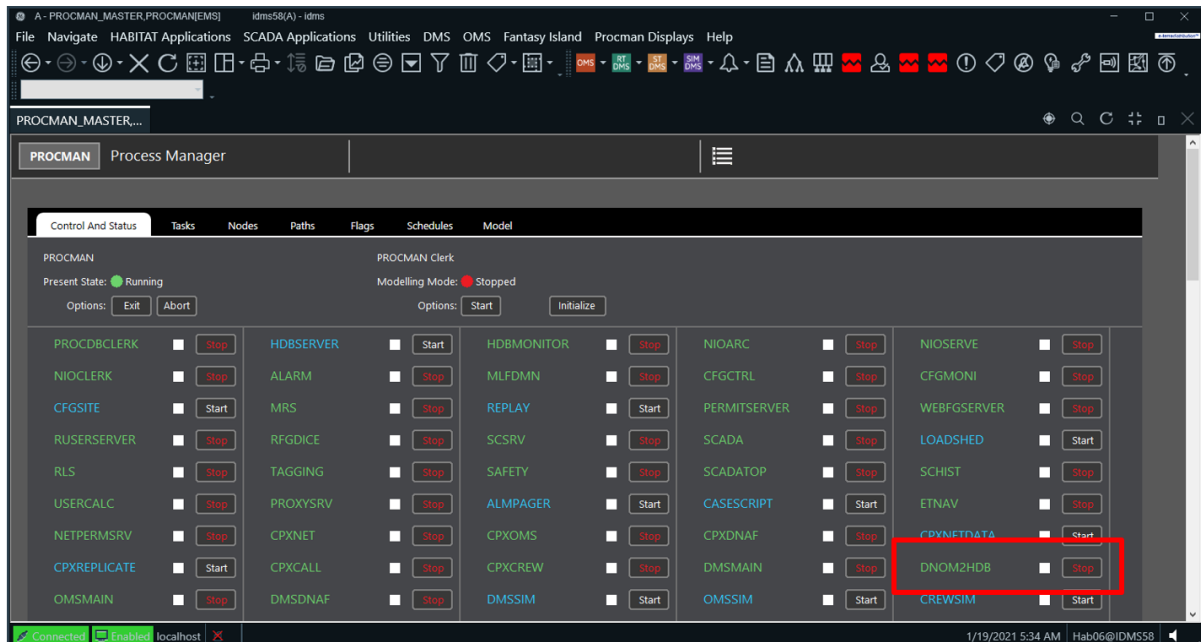
This chapter describes how to run DNOM2HDB and create queries.

The chapter also includes several examples on how to extract DNAF results. The database definition used in these examples is shown in [Database and Display Definition](#).

3.1. Starting DNOM2HDB

To start the DNOM2HDB application:

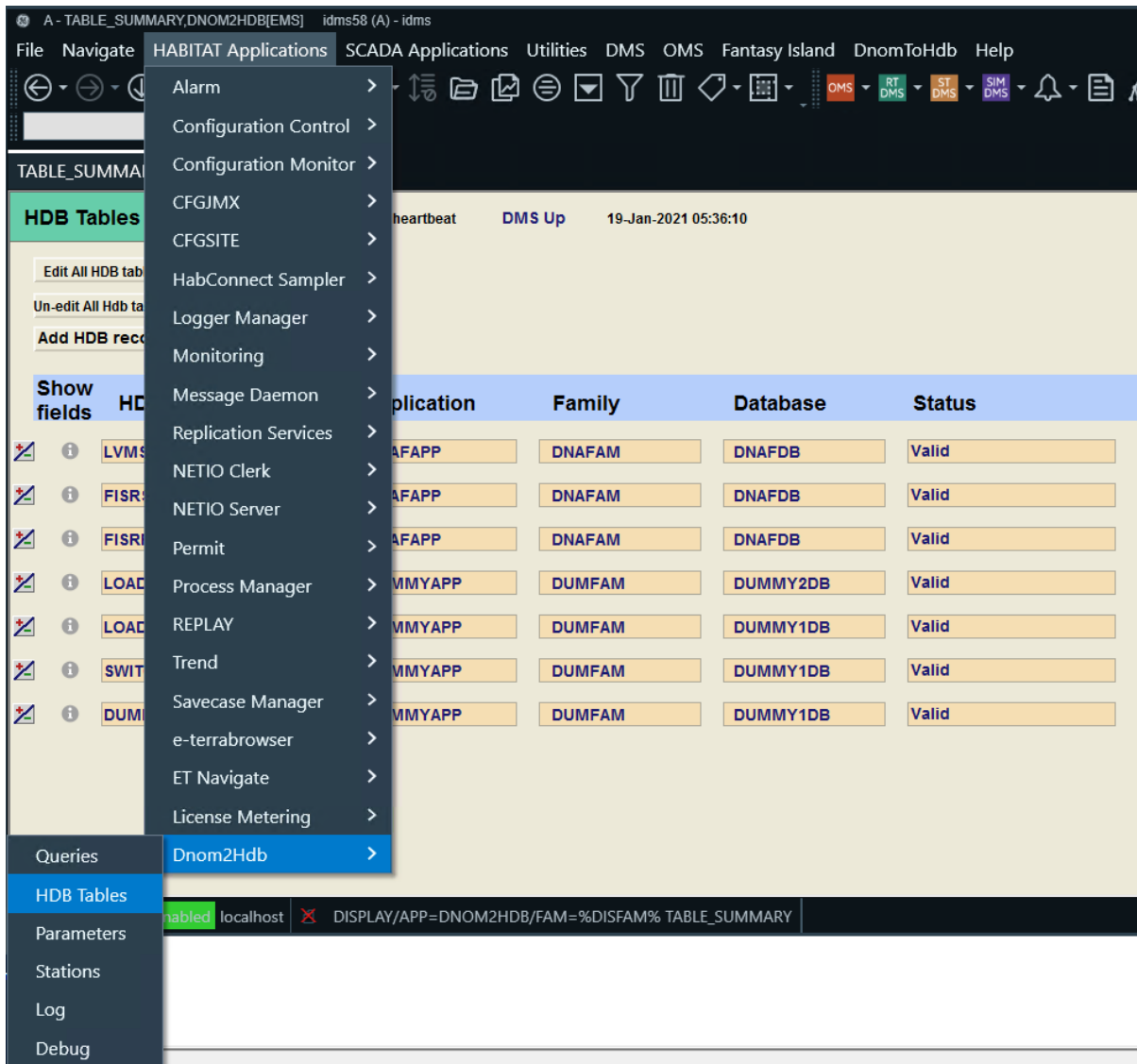
1. Using the Browser menu, navigate to HABITAT Applications > Process Manager > Process Manager Master, and click Start for the DNOM2HDB process.



Or: in the administrative command prompt, enter:

```
context dnom2hdb ems
dnom2hdb
```

2. Open Browser and access the DNOM2HDB displays via the main menu.



3.2. Query Helper Objects

DNOM2HDB copies data from DNOM fields that are associated with Query Helper objects into an Habitat database. This section describes the Query Helper objects that define DNAF application plans to be copied into an Habitat database.

For a high-level overview of the DNOM2HDB query syntax and execution triggers, refer to [Functional Overview](#).

Plans from FISR, AFR, LVM, FISRCA, and PLORS are defined via two Query Helper objects:

- OPTCONTROLPLAN
- OPTCONTROLMOVE

Plans can be filtered in OPTCONTROLPLAN according to DNAF types using the following

OPTCONTROLPLAN fields:

- SHOWAFRPLAN=TRUE
- SHOWFISRCAPLAN=TRUE
- SHOWPLORSPLAN=TRUE
- ISVVCPLAN=TRUE
- SHOWFISRPLAN=TRUE

OPTCONTROLPLAN and OPTCONTROLMOVE objects share a common field: PLANFUNCID.

Plans from FL (Fault Location) are defined via the PLANFARESULT Query helper object.

Results of PRV (Protection Validation) are defined in the SWT Query Helper object with the condition FADEVTYPE=TRUE.

The output status summary can be reached using the OPTCONTROLSOL Query helper object.

DNOM2HDB can automatically add the execution time value of the DNAF application and the station ID value to each row in a HABITAT clone.

An Historian FISR report can be created via the following three fields: DNAF execution time, station ID, and PLANFUNCID.

3.3. Configuring FISR and LVM Triggers

To add the FISR option:

1. From the DNOM2HDB Parameters display, right-click the "+/-" icon and select FISR.
2. In the Label Text field, enter any label name (for example, "FISR").
3. In the DNAF Status field, enter "Solved".

To add the LVM option:

1. From the DNOM2HDB Parameters display, right-click the "+/-" icon and select LVM.
2. In the Label Text field, enter any label name (for example, "LVM").
3. In the DNAF Status field, enter "Solved".

Parameters

☐ Freeze heartbeat
 DMS Up
 19-Jan-2021 05:43:41

Heartbeat
Seconds

DMS Server #1:

Host Name:

Socket:

Timeout: Seconds

DMS Server #2:

Host Name:

Socket:

Timeout: Seconds

Query menu for DNAF applications

	Label Text	DNAF Status
DPF	DPF	Solved / BLA Converged
FISR	FISR	Solved
LVM	LVM	Solved

- DPF
- AFR
- FISR
- FL
- LVM
- PRV
- Disable
- Show Related Displays...

3.4. Example 1: Extracting LVM Plans

To automatically copy the data defined in the ADMS LVM tabular displays into the HABITAT clone, copy the following fields of the OPTCONTROLMOVE DNOM object with respect to the where condition "ISVVCMOVE=TRUE":

- OPTMOVDEVNAME (Device Name)
- OPTMOVDEVID (Device Id)
- OPTMOVSUBNAME (Device Station)
- OPTMOVFEEDERNAME (Device Feeder)
- DEVICETYPE (Device Type)
- OPTMOVINITVALUE (Initial Position)
- OPTMOVFINALVALUE (Recommended Final Position)

Step Number	Energizing Station	Energizing Feeder	Device Name	Locate	Device Type	Control Type	Phase	Initial Position or	Recommended Final Position or	Plan Summary	Plan
1	BACHELOR	--	T1		Tap Changer	Raise/Lower	ABC	4	5		
2	BACHELOR	--	T2		Tap Changer	Raise/Lower	ABC	0	-1		
3	BACHELOR	F7932	3839		Capacitor	Open/Close	ABC	Open	Close		
4	BACHELOR	F7933	1650		Capacitor	Open/Close	ABC	Open	Close		
5	BACHELOR	F7935	4067		Capacitor	Open/Close	ABC	Open	Close		
6	BACHELOR	F7935	3749		Capacitor	Open/Close	ABC	Open	Close		
7	BACHELOR	F7935	1623		Capacitor	Open/Close	ABC	Open	Close		

This tabular display is defined in `vvcpplansteps_GridDisplay.xml`.

3.4.1. HDB Table Configuration

The data is copied in the LVMSTEP table `dnafdb.dbdef`.

To indicate which "HDB field" is written:

1. On the HDB Tables display, select the EDIT Mode check box.
2. Enter:
 - Record: LVMSTEP
 - Application: DNAFAPP
 - Family: DNAFAM
 - Database: DNAFDB
3. Click Add HDB Field to add eight HDB fields using Query Helper objects defined in [Example 1: Extracting LVM Plans](#).
4. Enter the list of HDB Field IDs, as shown below.

HDB record

☒ EDIT mode

Done

Record

LVMSTEP

Application

DNAFAPP

Family

DNAFAM

Database

DNAFDB

Keep rows at restart/failover

☐

Status

Edit

HDB field id	Type & size	Add HDB field
 OPTMOVDEVNAME	C * 20	
 OPTMOVDEVID	C * 20	
 OPTMOVSUBNAME	C * 15	
 OPTMOVFEEDERNAME	C * 10	
 DEVICETYPE	C * 15	
 OPTMOVINITVALUE	C * 10	
 OPTMOVFINALVALUE	C * 10	
 EXECTIME	T * 4	

5. Clear the EDIT Mode check box to validate the changes.

3.4.2. Query Configuration

To define the mapping between the fields of the DNOM object OPTCONTROLMOVE and the fields of the HDB record LVMSTEP:

1. On the Edit Query display, create and edit a Query.
2. Enter:
 - Query Name: LVM STEP (optional, any text can be used)
 - WHERE: ISVVCMOVE=TRUE (condition)
 - DNOM Object Type: OPTCONTROLMOVE

3. Select the following options:

- Selected Station List box
- Periodic option
- DNAF Applications option
- LVM check box

4. For each HDB Field ID, enter the corresponding DNOM Field and select the associated option button, as shown below.

Skip	Destination		Source				Static value
	HDB field id	Type & size	DNOM field	Trigger time	Present time	Constant value	
☐	OPTMOVDEVNAME	C * 20	OPTMOVDEVNAME	☐	☐	☐	☐
☐	OPTMOVDEVID	C * 20	OPTMOVDEVID	☐	☐	☐	☐
☐	OPTMOVSUBNAME	C * 15	OPTMOVSUBNAME	☐	☐	☐	☐
☐	OPTMOVFEEDERNAME	C * 10	OPTMOVFEEDERNAME	☐	☐	☐	☐
☐	DEVICETYPE	C * 15	DEVICETYPE	☐	☐	☐	☐
☐	OPTMOVINITVALUE	C * 10	OPTMOVINITVALUE	☐	☐	☐	☐
☐	OPTMOVFINALVALUE	C * 10	OPTMOVFINALVALUE	☐	☐	☐	☐
☐	EXEETIME	T* 4		☐	☐	☐	

5. Click the Trigger Time option button for EXEETIME.

6. Clear the EDIT Mode check box to validate the changes.

3.4.3. Verify Results

DNAF_LVMSTEP (dnaf_lvmstep.fgdisplay) is a display defined in the dnaf_dset.ddl file. When LVM is

solved for a station, the LVM results must appear in this display.

1. Call up a Browser viewport and enter in its header entry box: "DNAFAPP/DNAF_LVMSTEP".

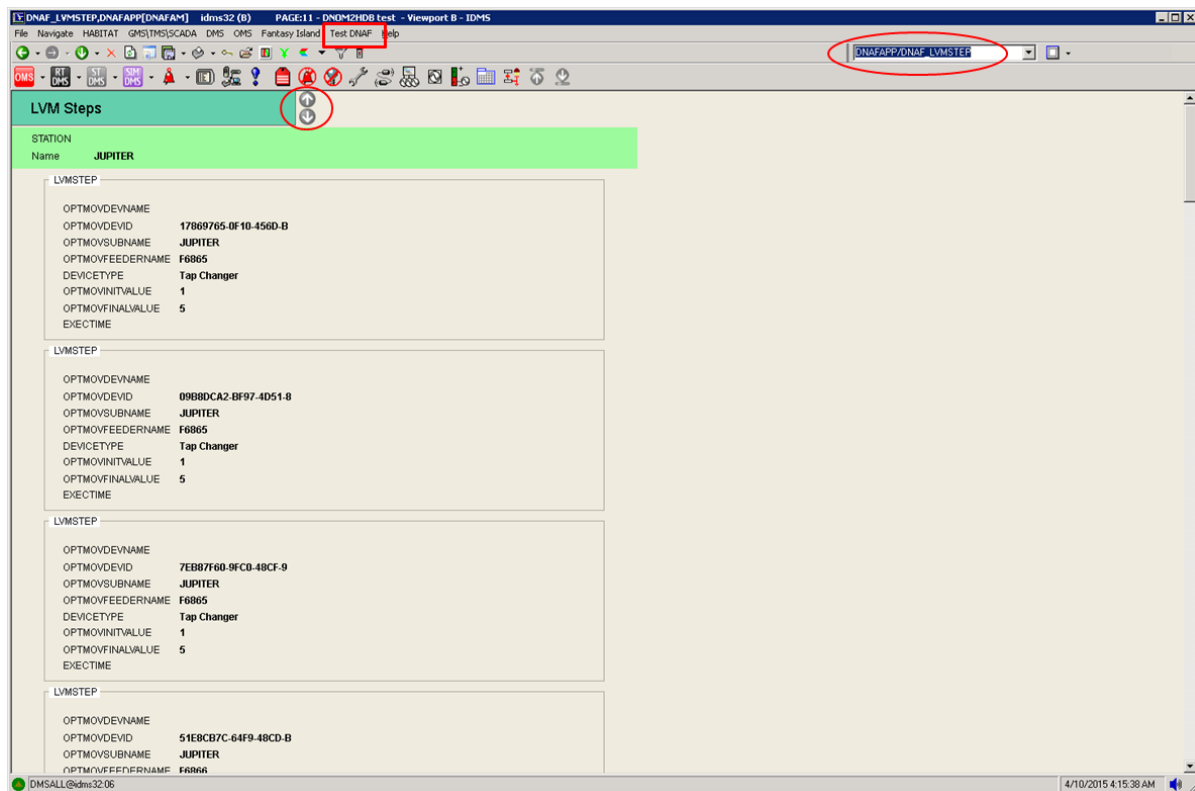
Note

The files that define the DNAF displays (dnaf_lvmstep.fgdisplay, dnaf_firstep.fgdisplay, and dnaf_firplan.fgdisplay) must be located in the %HABITAT_WEBROOT%\displays\dnafapp directory.

Note

To allow calling the DNAF displays, make sure that the DNAFAPP application belongs to the DNAFAM family and has the required permissions. For more information, refer to the *DNOM2HDB Installation Guide*. In addition, refer to [Creating an HDB Clone](#).

2. Select a station and watch the data as shown below.



3. The result can be verified by comparing with the ADMS tabular display, as shown below:

JUPITER.VVPLANSTEPS_GRIDDISPLAY_RT - Viewport D - IDMS

File Navigate HABITAT GMS/TMS/SCADA DMS OMS Fantasy Island Help

JUPITER - Load and Volt/VAR Management (Execution Time: 4/9/2015 8:37:02 AM Solution Time: 4/9/2015 8:36:56 AM)

Device Name	Device Id	Device Station	Device Feeder	Device Type	Phase	Locate	Step Number	Initial Position	Recommended Final Position	Plan Summary	Plan Statistics
17869765-0F1...		JUPITER	F6865	Tap Changer	C		1	1	5		
09B8DCA2-BF...		JUPITER	F6865	Tap Changer	B		2	1	5		
7EB87F60-9F...		JUPITER	F6865	Tap Changer	A		3	1	5		
51E8CB7C-64...		JUPITER	F6866	Tap Changer	C		4	0	4		
69340590-B74...		JUPITER	F6866	Tap Changer	B		5	0	4		
RCF6RDR4-F4		JUPITER	F6866	Tap Changer	A		6	0	4		

4/10/2015 4:21:03 AM

Note the LVM status, solution time, and device ID for a station on these displays below.

Real-Time DNAF Solution - COHO / 22509596

Application: Load and Volt/VAR Management

Station	In	Out	Status	Evaluation Time	DPF	PRV	LVM	FISR	AFR	FL
JUPITER			Solved	4/9/2015 8:36:...						

STATION_SUMMARY.DNOM2HDB[EMS] idms32 (C) - Viewport C - IDMS

File Navigate HABITAT GMS/TMS/SCADA DMS OMS Fantasy Island Dnom2Hdb Help

Selected Station list ☐ Freeze heartbeat DMS Up 10-Apr-2015 04:26:16

☐ Edit the station selection

Station		DPF	AFR	FISR
✓ JUPITER	2015-04-10 04:10:32	Not Solved	Not Solved	
Suspect - BLA Not Converged		Unsolved	Unsolved	
LVM	2015-04-09 08:36:56	Not Solved	Topology	10-Apr-2015 03:19:58
Solved		Unsolved		

LVM Steps

STATION

Name **JUPITER**

LVMSTEP

OPTMOVDEVNAME

OPTMOVDEVID **17869765-0F10-456D-B**

OPTMOVSUBNAME **JUPITER**

OPTMOVFEEDERNAME **F6865**

DEVICETYPE **Tap Changer**

OPTMOVINITVALUE **1**

OPTMOVFINALVALUE **5**

EXECTIME

3.5. Example 2: Extracting FISR Plans

The objective of the following plans is to automatically copy the data defined in the ADMS tabular displays below into the HABITAT clone:

RT - Viewport G - IDMS

File Navigate HABITAT GMS|TMS|SCADA DMS OMS Fantasy Island Help

</

This display is based on two tabular displays: `FISRPlans_GridDisplay.xml` and `switchingplancontrolmoves_GridDisplay.xml`. The second tabular display appears after clicking the Show Steps field on a FISR Plan row.

Option 1: Copy the following fields of the OPTCONTROLPLAN DNOM object with respect to the condition "SHOWFISRPLAN=TRUE":

- SOLUTIONID
- PLANFUNCID
- NUMRANK (Rank)
- NUMCONTROLMOVE (Number of Moves)
- NMCSTNRT (Number of Customers Not Restored)
- KWNTRST (Unserved kW)
- MAXSEGLD (Maximum Segment Loading (%))
- MAXLDVLT (Maximum Load Voltage)
- MINLDVLT (Minimum Load Voltage)
- PROTVAL (Protection Validation)

Option2: Copy the following fields of the OPTCONTROLMOVE DNOM object:

- PLANFUNCID
- PLANSTEPNO (Step Number)
- DEVICETYPE (Device Type)
- DEVICENAME (Device Name)
- OPTMOVDEVID (Device ID)
- SUBNAME (Device Substation)
- OPTMOVINITVALUE (Initial State)
- OPTMOVFINALVALUE (Recommended State)

Note

PLANFUNCID is a shared field value. This way, it is possible to filter the steps that belong to a FISR Plan.

3.5.1. HDB Table Configuration

The data is copied in the FISRPLAN and FISRSTEP records of `dnafdb.dbdef`.

1. On the HDB table display, indicate the HDB field to be written.
2. Add the following two HDB tables and edit them:

For "FISRPLAN" HDB Record, enter:

- Record: FISRPLAN
- Application: DNAFAPP
- Family: DNAFAM
- Database: DNAFDB

3. Click Add HDB Field and enter the list of HDB field ids using Query Helper objects defined in [Example 2: Extracting FISR Plans](#).

The screenshot shows a window titled "TABLE_EDITION, DNOM2HDB[EMS] idms32 (TABLE_EDITION) - Viewport TABLE_EDITION - IDMS". The window has a tab labeled "HDB record" and a checkbox for "EDIT mode" which is currently unchecked. A "Done" button is in the top right corner. Below the tab, there are several fields for configuration:

- Record: FISRPLAN
- Application: DNAFAPP
- Family: DNAFAM
- Database: DNAFDB
- Keep rows at restart/failover: ☐
- Status: Running

Below these fields is a table with two columns: "HDB field id" and "Type & size".

HDB field id	Type & size
SOLUTIONID	C * 30
PLANFUNCID	C * 40
NUMRANK	I * 4
NUMCONTROLMOVE	I * 4
NMCSTNRT	I * 4
KWNTRST	C * 15
MAXSEGLD	C * 15
MAXLDVLT	C * 20
MINLDVLT	C * 20
PROTVL	C * 15
EXEETIME	T * 4

4. Clear the EDIT Mode check box to validate the changes.

5. For the "FISRSTEP" HDB record, enter:

- Record: FISRSTEP
- Application: DNAFAPP
- Family: DNAFAM
- Database: DNAFDB

6. Click Add HDB Field and enter the list of HDB field ids, as shown below.

TABLE_EDITION_DNOM2HDB[EMS] idms32 (TABLE_EDITION) - Viewport TABLE_EDITION - IDMS

HDB record ☐ EDIT mode Done

Record:

Application:

Family:

Database:

Keep rows at restart/failover: ☐

Status:

HDB field id	Type & size
PLANFUNCID	C * 40
PLANSTEPNO	I * 4
DEVICETYPE	C * 15
DEVICENAME	C * 20
OPTMOVDEVID	C * 20
SUBNAME	C * 15
OPTMOVINITVALUE	C * 10
OPTMOVFINALVALUE	C * 10
EXECTIME	T * 4

7. Clear the EDIT Mode check box to validate the changes.

3.5.2. Query Configuration

Define the mapping between the following fields:

1. The fields of the DNOM object OPTCONTROLPLAN and the fields of the HDB record FISRPLAN.
2. The fields of the DNOM object OPTCONTROLMOVE and the fields of the HDB record FISRSTEP.

Add two HDB tables and edit them on the HDB Tables display.

1. For "FISR PLAN" Query, enter:
 - Query Name: FISR PLAN (optional, any text is allowed)
 - WHERE: SHOWFISRPLAN=TRUE (condition)
 - DNOM Object Type: OPTCONTROLPLAN
2. Select the following options:
 - Selected Station List box
 - Periodic option
 - DNAF Applications option
 - FISR check box
3. For each HDB field id, enter the corresponding DNOM field and select the associated option as shown below.

QUERY_EDITION, DNOM2HDB[EMS] idms32 (QUERY_EDITION) - Viewport QUERY_EDITION - IDMS

Edit QUERY ☐ EDIT mode Status **Run as Timer** Last execution time for this query 10-Apr-2015 03:20:51 Done

Query Name **FISR PLAN**

FROM ☒ Selected station list Number max of stations per DNOM query (0 means no limit) **0**
☐ Custom entered stations

WHERE (condition) **SHOWFISRPLAN=TRUE**

DNOM Object Type **OPTCONTROLPLAN**

HDB Record Name **FISRPLAN** ⓘ

Trigger ☐ Disable
☐ On demand
☒ Periodic **60** seconds ☐ Start time ☐ Stop time
☐ No extra constraint ☐ DPF
☒ DNAF applications ☒ FISR
☐ Topology change ☐ LVM

Skip	Destination		Source				Static value
	HDB field id	Type & size	DNOM field	Trigger time	Present time	Constant value	
☐	SOLUTIONID	C * 30	☒ SOLUTIONID	☐	☐	☐	☐
☐	PLANFUNCID	C * 40	☒ PLANFUNCID	☐	☐	☐	☐
☐	NUMRANK	I * 4	☒ NUMRANK	☐	☐	☐	☐
☐	NUMCONTROLMOVE	I * 4	☒ NUMCONTROLMOVE	☐	☐	☐	☐
☐	NMCSTNRT	I * 4	☒ NMCSTNRT	☐	☐	☐	☐
☐	KWNTRST	C * 15	☒ KWNTRST	☐	☐	☐	☐
☐	MAXSEGLD	C * 15	☒ MAXSEGLD	☐	☐	☐	☐
☐	MAXLDVLT	C * 20	☒ MAXLDVLT	☐	☐	☐	☐
☐	MINLDVLT	C * 20	☒ MINLDVLT	☐	☐	☐	☐

4. Select the Trigger Time option button for EXECTIME.
5. Clear the EDIT Mode check box to validate the changes.
6. For FISR STEP query, enter:
 - Query Name: FISR STEP (any text is allowed)
 - DNOM Object Type: OPTCONTROLMOVE
7. Select the following options:
 - Selected Station List box
 - Periodic option
 - DNAF Applications option
 - FISR check box
8. For each HDB field id, enter the corresponding DNOM field and select the associated option button as shown below.

QUERY_EDITION,DNOM2HDB[EMS] idms32 (QUERY_EDITION) - Viewport QUERY_EDITION - IDMS

Edit QUERY ☐ EDIT mode Status **Run as Timer** Last execution time for this query **10-Apr-2015 03:22:00** **Done**

Query Name **FISR STEP**

FROM ☒ Selected station list Number max of stations per DNOM query (0 means no limit) **0**
☐ Custom entered stations

WHERE (condition)

DNOM Object Type **OPTCONTROLMOVE**

HDB Record Name **FISRSTEP** ⓘ

Trigger ☐ Disable
☐ On demand
☒ Periodic **60** seconds ☐ Start time ☐ Stop time
☐ No extra constraint ☐ DPF
☒ DNAF applications ☒ FISR
☐ Topology change ☐ LVM

Skip	Destination		Source				Static value
	HDB field id	Type & size	DNOM field	Trigger time	Present time	Constant value	
☐	PLANFUNCID	C * 40	☒ PLANFUNCID	☐	☐	☐	☐
☐	PLANSTEPNO	I * 4	☒ PLANSTEPNO	☐	☐	☐	☐
☐	DEVICETYPE	C * 15	☒ DEVICETYPE	☐	☐	☐	☐
☐	DEVICENAME	C * 20	☒ DEVICENAME	☐	☐	☐	☐
☐	OPTMOVDEVID	C * 20	☒ OPTMOVDEVID	☐	☐	☐	☐
☐	SUBNAME	C * 15	☒ SUBNAME	☐	☐	☐	☐
☐	OPTMOVINITVALUE	C * 10	☒ OPTMOVINITVALUE	☐	☐	☐	☐
☐	OPTMOVFINALVALUE	C * 10	☒ OPTMOVFINALVALUE	☐	☐	☐	☐
☐	EXEETIME	T * 4	☐	☒	☐	☐	☐

9. Select the Trigger Time option button for EXEETIME.

3.5.3. Verify Result

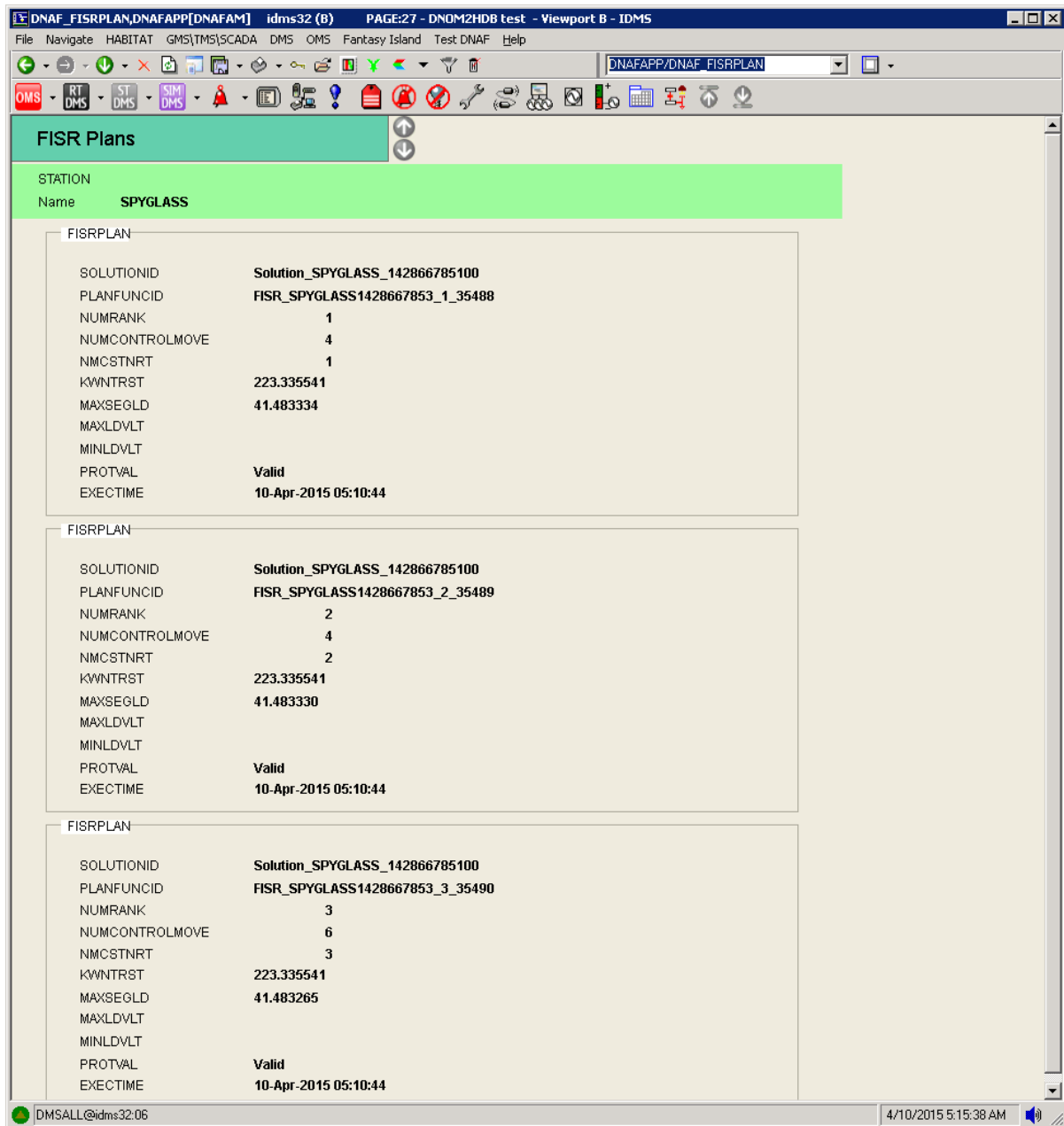
The DNAF_FISRPLAN (dnaf_fisrplan.fgdisplay) display is defined in the dnaf_dset.ddl file. After FISR is solved for a station, the FISR results must appear in this display.

1. Call up a Browser viewport and enter in the header entry box: "DNAFAPP/DNAF_FISRPLAN".

Note

To allow calling this display, make sure that the DNAFAPP application belongs to the DNAFAM family and has the required permissions. For more information, see [Creating an HDB Clone](#).

2. From this display, select a station and watch the FISR data (see the image below).



- Click the text related to PLANFUNCID to view the associated Move Control Steps of a FISR Plan (see the image below).

FISR Plans	
STATION	
Name	SPYGLASS
FISRPLAN	
SOLUTIONID	Solution_SPYGLASS_142866785100
PLANFUNCID	FISR_SPYGLASS1428667853_1_35488
NUMRANK	1
NUMCONTROLMOVE	4
NMCSTNRT	1

Click to get STEPs

After clicking the text, the following display appears:

FISR Steps	
FISRSTEP	
PLANFUNCID	FISR_SPYGLASS1428667853_1_35488
PLANSTEPNO	1
DEVICETYPE	Switch
DEVICENAME	11683
OPTMOVDEVID	34810620
SUBNAME	SPYGLASS
OPTMOVINITVALUE	Close
OPTMOVFINALVALUE	Open
EXECTIME	10-Apr-2015 05:10:44
FISRSTEP	
PLANFUNCID	FISR_SPYGLASS1428667853_1_35488
PLANSTEPNO	2
DEVICETYPE	Breaker
DEVICENAME	6464F
OPTMOVDEVID	E90351C4-31D7-405C-B
SUBNAME	SPYGLASS
OPTMOVINITVALUE	Open
OPTMOVFINALVALUE	Close
EXECTIME	10-Apr-2015 05:10:44
FISRSTEP	

This display filters the FISR steps based on filtering on the PLANFUNCID value.

4. Database and Display Definitions

This chapter presents the database definitions and displays used throughout [Running DNOM2HDB](#).

4.1. Example dth_test_dnaf.dbef

The dth_test_dnaf.dbef database is used as an example test case for the DNOM2HDB application. By default, this file is located in D:\Eterra\habsrc06\habuser\special\Dnom2hdb.

The dth_test_dnaf.dbef contains the following database definitions:

```
//DBDEF  DNAFDB TITLE= (DNAFDB) FLAGS=(VALID)
//RECTYP FISRPLAN(500) PARENT=STATION
//  FLAGS=(HIER)
//RECTYP FISRSTEP(5000) PARENT=STATION
//          FLAGS=(HIER)
//RECTYP LVMSTEP(1000) PARENT=STATION
//          FLAGS=(HIER)
-----
//RECTYP STATION(200) NOPARENT
//  FLAGS=(GLOBAL, HIER)
//FLDPAR DUMPAR FLAGS=()
//FIELD  NAME_STATION C*15 FILL=20
-----

//FIELD SOLUTIONID_FISRPLAN C*30 FILL=20
//FIELD PLANFUNCID_FISRPLAN C*40 FILL=20
//FIELD NUMRANK_FISRPLAN I*4 FILL=00
//FIELD NUMCONTROLMOVE_FISRPLAN I*4 FILL=00
//FIELD NMCSTNRT_FISRPLAN I*4 FILL=00
//FIELD KWNTRST_FISRPLAN C*15 FILL=20
//FIELD MAXSEGLD_FISRPLAN C*15 FILL=20
//FIELD MAXLDVLT_FISRPLAN C*20 FILL=20
//FIELD MINLDVLT_FISRPLAN C*20 FILL=20
//FIELD PROTVAL_FISRPLAN C*15 FILL=20
//FIELD EXECTIME_FISRPLAN T*4 FILL=00

//FIELD PLANFUNCID_FISRSTEP C*40 FILL=20
//FIELD PLANSTEPNO_FISRSTEP I*4 FILL=00
//FIELD DEVICETYPE_FISRSTEP C*15 FILL=20
//FIELD DEVICENAME_FISRSTEP C*20 FILL=20
//FIELD OPTMOVDEVID_FISRSTEP C*20 FILL=20
//FIELD SUBNAME_FISRSTEP C*15 FILL=20
//FIELD OPTMOVINITVALUE_FISRSTEP C*10 FILL=20
//FIELD OPTMOVFINALVALUE_FISRSTEP C*10 FILL=20
//FIELD EXECTIME_FISRSTEP T*4 FILL=00

//FIELD PLANFUNCID_LVMSTEP C*40 FILL=20
//FIELD OPTMOVDEVNAME_LVMSTEP C*20 FILL=20
//FIELD OPTMOVDEVID_LVMSTEP C*20 FILL=20
```

```
//FIELD OPTMOVSUBNAME_LVMSTEP C*15 FILL=20
//FIELD OPTMOVFEEDERNAME_LVMSTEP C*10 FILL=20
//FIELD DEVICETYPE_LVMSTEP C*15 FILL=20
//FIELD OPTMOVINITVALUE_LVMSTEP C*10 FILL=20
//FIELD OPTMOVFINALVALUE_LVMSTEP C*10 FILL=20
//FIELD EXECTIME_LVMSTEP T*4 FILL=00
```

4.2. Creating an HDB Clone

1. Enter the following commands at a command prompt:

```
loadhab dth_test_dnaf.dbdef
loadhab dth_test_dnaf.cls
hdbcloner -c create_clone -f dnafam -a dnafapp
```

The displays for testing the examples are defined in the dth_test_dnaf_dset.ddl and dth_test_dnaf_eset.ddl files. The default location is D:\Eterra\habdata06\uidbroot\ddl.s.

2. Set the following permissions to allow users to open these displays (under the DNAFAPP application and DNAFAM family). For more information about modeling the DNAFAPP mode, refer to the *DNOM2HDB Installation Guide*.

Applications for Mode		DNAMEPLT			
Application	Family	Read	Write	Execute	Audible
ALARM	EMS	☒	☐	☐	☐
ALMPAGER		☐	☐	☐	☐
DNOM2HDB	EMS	☒	☒	☒	☒
CASESRV	HABITAT	☒	☒	☒	☒
DNAFAPP	DNAFAM	☒	☒	☒	☒